# Predictive Modeling of Cardiovascular Risk using Random Forests

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## 1. Abstract and Methodology

In this paper, we develop a supervised machine learning benchmark for clinical cardiovascular risk assessment. We train an ensemble random forest classifier with 100 trees, max_depth of 6, and a test_size of 0.25. The random seed is set to 42 for reproducible evaluation.

## 2. Results and Verification

Across 5-fold cross-validation, our proposed random forest reaches an accuracy of 93.3% and an F1 score of 92.5% on the test cohort.